

Lec 01

Probability and Statistics for computer Science

- Links with other courses

- Data science
- Machine Learning
- Simulations
- Robotics
- Engineering

- Contents

- Statistics

- Definitions
- Types and presentation of data
- Measure of central tendency & dispersion
- Curve fitting (future predictions)

- Probability

- Basics (sets, definitions, laws)
- Counting
- Random variables
- Distributions
- Misc... (joint probabilities etc)

- Applications

$\propto$ Entropy

- Monte carlo simulations
- Markov chains imp $\rightarrow$ reinforcement learning
- Machine learning
 - Spam detection
 - Robot localization